

Classifying Quadrilaterals from Defining Properties

Answer Key

High School Geometry Polygons

Quick Answers

- | | |
|----------|-----------------------------------|
| 1. 15 | 5. 16 |
| 2. 40 | 6. $\frac{1}{2}$, -2, 4.47, 4.47 |
| 3. 6 | 7. — |
| 4. 5, 14 | 8. — |
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Curriculum

Level: High School Geometry Polygons

HSG-CO.A–HSG-CO.D – *problems 1, 2, 3, 4, 5, 7, 8*

HSG-GPE.B.4–B.7 – *problem 6*

Difficulty 1–4 of 5 across 12 tagged checks.

1. Both pairs of opposite sides parallel.

Step 1. Classify. Two pairs of parallel sides is the definition of a **parallelogram**. Nothing marks the sides congruent and nothing marks a right angle, so parallelogram is as specific as the marks allow — not rhombus, not rectangle.

Step 2. Pick the property. In a parallelogram opposite angles are congruent, and $\angle A$ and $\angle C$ are opposite.

Step 3. Solve. $5x = 75$, so dividing by 5 gives

$x = 15$

Step 4. Check. $5(15) = 75$, which matches $m\angle C$ ✓

Common wrong answer: setting $5x + 75 = 180$ (giving $x = 21$) uses the *consecutive*-angle property on a pair of *opposite* angles. Ask which two angles share a side.

2. No marks at all.

Step 1. Classify. With no congruence and no parallel marks, nothing forces any special family. The only honest name is *quadrilateral*: it may happen to be a trapezoid, but the figure does not say so, and a name has to be forced by the given properties, not guessed from the drawing.

Step 2. Pick the property. One angle fact holds for every quadrilateral: the four interior angles sum to 360° .

Step 3. Solve. $(x + 20) + 2x + (3x - 10) + (2x + 30) = 360$, so $8x + 40 = 360$ and $8x = 320$.

$x = 40$

Step 4. Check. The four angles are 60° , 80° , 110° and 110° , and they total 360° ✓

Teaching note: no pair of these angles sums to 180° , which confirms there are no parallel sides — the figure really is a plain quadrilateral.

3. Four congruent sides plus one right angle.

Step 1. Classify. Four congruent sides makes it a rhombus; a rhombus with a right angle is a **square**. (One right angle is enough: in a parallelogram consecutive angles are supplementary, so one 90° angle forces all four.)

Step 2. Pick the property. All four sides of a square are congruent, so the two labelled sides are equal.

Step 3. Solve. $3x - 5 = x + 7$, so $2x = 12$.

$$x = 6$$

Step 4. Check. $3(6) - 5 = 13$ and $6 + 7 = 13$ ✓ — every side is 13 units.

Common wrong answer: “rhombus”. The marks force the more specific name, and the directions ask for the most specific one.

4. Diagonals that bisect each other and are congruent.

Step 1. Classify. Diagonals that bisect each other force a parallelogram. Adding *congruent* diagonals forces a **rectangle** — that is the diagonal test for a rectangle. Nothing says the diagonals are perpendicular, so it need not be a square.

Step 2. Pick the property. Congruent diagonals means $JL = KM$.

Step 3. Solve. $4x - 6 = x + 9$, so $3x = 15$.

$$x = 5$$

Step 4. Report the length. $JL = 4(5) - 6 = 20 - 6$.

$$JL = 14$$

Step 5. Check. $KM = 5 + 9 = 14$ as well ✓

Listen for: “the picture looks like a rectangle, so I did not need the diagonals.” The figure is labelled not to scale on purpose — the reasoning has to come from the two diagonal facts.

5. Congruent sides with perpendicular bisecting diagonals.

Step 1. Classify. Four congruent sides gives a rhombus, and the diagonals of a rhombus are exactly the ones that bisect each other at right angles — consistent, and no right angle is marked at any *vertex*, so **rhombus** is the most specific name.

Step 2. Pick the property. The right angle at E puts $\angle BAE$, $\angle ABE$ and the 90° angle in one triangle, ABE , and a triangle’s angles sum to 180° .

Step 3. Solve. $90 + (4x - 6) + 2x = 180$, so $6x + 84 = 180$ and $6x = 96$.

$$x = 16$$

Step 4. Check. $m\angle ABE = 4(16) - 6 = 58^\circ$ and $m\angle BAE = 2(16) = 32^\circ$; $90 + 58 + 32 = 180$ ✓

Teaching note: the perpendicular diagonals are what turn a quadrilateral problem into a right-triangle problem. That move — cut the quadrilateral into triangles with a diagonal — is worth naming out loud.

6. Classifying $ABCD$ from its vertices.

Step 1. Plan. Slopes decide parallel and perpendicular; distances decide congruent. Compute both for two adjacent sides.

Step 2. Slopes. For $A(-2, 1)$ and $B(2, 3)$: $\frac{3-1}{2-(-2)} = \frac{2}{4}$, so the slope of $\overline{AB}$ is

$$\frac{1}{2}$$

For $B(2, 3)$ and $C(4, -1)$: $\frac{-1-3}{4-2} = \frac{-4}{2}$, so the slope of $\overline{BC}$ is

$$-2$$

Step 3. What the slopes force. The product is $\frac{1}{2} \cdot (-2) = -1$, so $\overline{AB} \perp \overline{BC}$: the angle at B is right. Running the same computation on the other two sides gives slope $\frac{1}{2}$ for $\overline{DC}$ and -2 for $\overline{AD}$, so both pairs of opposite sides are parallel and all four angles are right — a rectangle so far.

Step 4. Lengths. $AB = \sqrt{4^2 + 2^2} = \sqrt{20}$, so

$$AB \approx 4.47$$

$BC = \sqrt{2^2 + (-4)^2} = \sqrt{20}$, so

$$BC \approx 4.47$$

Step 5. Name it. Adjacent sides are congruent as well as perpendicular, so the rectangle is in fact a **square**. Perpendicular adjacent sides give the right angles; equal adjacent sides give the equal sides.

Common wrong answer: stopping at “rectangle”. The lengths are what promote it, and the directions ask for the most specific name.

7. Is a rhombus a rectangle? *Open justification — flagged for manual review. The response below is a model full-credit answer.*

Step 1. Answer. *Sometimes* — and only in one case.

Step 2. The separating property. A rhombus is a parallelogram with four congruent sides; a rectangle is a parallelogram with four right angles. Neither definition mentions the other’s condition, so being a parallelogram is all they are guaranteed to share. The rhombus $WXYZ$ on the sheet has four congruent sides and no right angle, which is a counterexample on its own.

Step 3. When it is true. A quadrilateral that satisfies both definitions — four congruent sides *and* four right angles — is a square, and a square is both a rhombus and a rectangle. So “sometimes”, with the overlap being exactly the squares.

Step 4. The marking. On the figure the student should add a right-angle square at one vertex (equivalently, mark the diagonals congruent). Either addition promotes the rhombus to a square, which is the only rhombus that is also a rectangle.

Full credit: the word *sometimes*, a reason that names congruent sides against right angles, and a mark that would force a right angle. Answering “never” is a common slip — it forgets the square.

8. One pair congruent and parallel. *Open written argument — flagged for manual review. Model answer below.*

Step 1. Set up. Given: $\overline{QR} \cong \overline{TS}$ and $\overline{QR} \parallel \overline{TS}$. Draw the diagonal $\overline{QS}$; it cuts the quadrilateral into $\triangle QRS$ and $\triangle STQ$.

Step 2. The argument. $\angle RQS \cong \angle TSQ$, because they are alternate interior angles on the parallel sides $\overline{QR}$ and $\overline{TS}$ with transversal $\overline{QS}$. Also $\overline{QS} \cong \overline{SQ}$ (reflexive), and $\overline{QR} \cong \overline{TS}$ is given. By SAS, $\triangle QRS \cong \triangle STQ$.

Step 3. Finish. CPCTC gives $\overline{RS} \cong \overline{QT}$, so both pairs of opposite sides are congruent, and a quadrilateral with both pairs of opposite sides congruent is a parallelogram. (The named shortcut for the whole argument: *if one pair of opposite sides is both congruent and parallel, the quadrilateral is a parallelogram.*)

Step 4. Rectangle? No. Nothing in the given marks constrains any angle, and the drawing itself shows a slanted parallelogram with no right angle. A rectangle would additionally need a right angle, or congruent diagonals.

Full credit: a chain that reaches parallelogram with a reason named at each step, plus a “no” on the rectangle question backed by what is missing. Asserting the parallelogram conclusion with no reasons is the common shortfall.